

Technology is evolving continuously today, and its cycle time is hardly a couple of years. Most technologies become obsolete and are taken over by newer and better technologies in this short span of time. Organisations need to keep pace with the latest technology innovations to sustain their business. There are many programming languages, frameworks, tools and data management approaches available in the market and choosing the right set to develop an application is extremely important. Right choices lead to a better application and ultimately better business results.

Strategy to build a great app

Choosing the right strategy and approach to build a great app is important. Here are the key considerations.

Find the business need

Defining the purpose and the need of an app for the market or business is the most important. Some primary and secondary research must be done to understand the macro and micro economic factors, the market players and the competition, and to also plan the financial viability of building a successful app.

Choose your technology stack

- Maintain a list of technologies in the following categories:
 - Persist (Continue using)
 - Adopt (Add to your technology stack)
 - Trial (Experiment with them)
 - Refrain (Not recommended for use)

Define your SECURED architectural principles

- Security
 - Follow the least privilege principle, which mitigates risk patterns
- Efficiency

- Consumes fewer resources, good performance, automated processes
- Consistency
 - Follows repeatable patterns
- Usability
 - Intuitive in usage
- Resiliency
 - Withstands failures
- Enterprise-value driven
 - Customer-centric, delivers business value
- Disposability
 - Easy to modify, update and replace

Choose your development method

Depending on whether you are building an app for a single platform or multiple platforms, you can choose:

- Native app development
- Hybrid app development
- Cross-platform app development
- Progressive web app development

If you want your application to run on both desktops and mobile devices, and automatically adjust for the different screen sizes and view ports, you can use a responsive web design.

Choose which type of app you want to build

Depending on the need, one can build a web application or a mobile application. A web application runs on a web server and is accessible using a browser. It can be accessed on both desktops and mobile devices but requires an internet connection. A mobile app is a piece of software that is installed on a mobile device like smartphones, tablets, smart watches, etc.

Web applications are much easier to build to reach a large audience as they can run on all types of operating systems as long as the browsers support them. On the other hand, mobile apps

typically give great user experience and have advanced in-built features like a camera, GPS, etc.

Key design considerations for building a great app

The following are key aspects to consider if you want to build a great app.

Build a strong core component

- All great apps have strong core features, which serve as the nucleus of the app.

Separate the business and technical components

- Keep the business logic separate from the tech components, and build with environment awareness.
- Great apps run well in all environments, be it development, testing, staging or production.

Include the configuration approach

- Configure all the possible dynamic variables to pass during runtime.

Design for failure

- Include retry and fail-safe mechanisms in your application.

Never lose track of user-flow and user-journey

- Clearly define, design and develop for all the possible user journeys in your app.

Never underestimate user experience

- A great app will be inherently intuitive and seamless to use.

Include auditing and logging mechanisms

- All great apps will have audit and logging for debugging purposes.

It is often said that “What you cannot measure, you cannot improve.” Having a measurement framework for your app and collecting metrics is extremely important. App diagnostic tools can be used for this. Also, the development, testing and deployment processes should be automated. With cloud based development, this can be achieved very easily. **END** 

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